

Tables and PERFORM..VARYING

TaxTotal



Variable = Named location in memory

PAYENum

CountyNum

TaxPaid



PROCEDURE DIVISION.

Begin.

OPEN INPUT TaxFile

READ TaxFile

AT END SET EndOfTaxFile TO TRUE

END-READ

PERFORM UNTIL EndOfTaxFile

ADD TaxPaid TO TaxTotal

READ TaxFile

AT END SET EndOfTaxFile TO TRUE

END-READ

END-PERFORM.

DISPLAY "Total taxes are ", TaxTotal

CLOSE TaxFile

STOP RUN.

The program
to calculate
the total taxes
paid for the
country is
easy to write.

BUT.

What do we do
if we want to
calculate the
taxes paid in
each **county**?

County1
TaxTotal

County2
TaxTotal

County3
TaxTotal

County4
TaxTotal

County5
TaxTotal



PROCEDURE DIVISION.

Begin.

OPEN INPUT TaxFile

READ TaxFile

AT END SET EndOfTaxFile TO TRUE

END-READ

PERFORM SumCountyTaxes UNTIL EndOfTaxFile

DISPLAY "County 1 total is ", County1TaxTotal

: : : 24 Statements : : :

DISPLAY "County 26 total is ", County26TaxTotal

CLOSE TaxFile

STOP RUN.

SumCountyTaxes.

IF CountyNum = 1 ADD TaxPaid TO County1TaxTotal

END-IF

: : : 24 Statements : : :

IF CountyNum = 26 ADD TaxPaid TO County26TaxTotal

END-IF

READ TaxFile

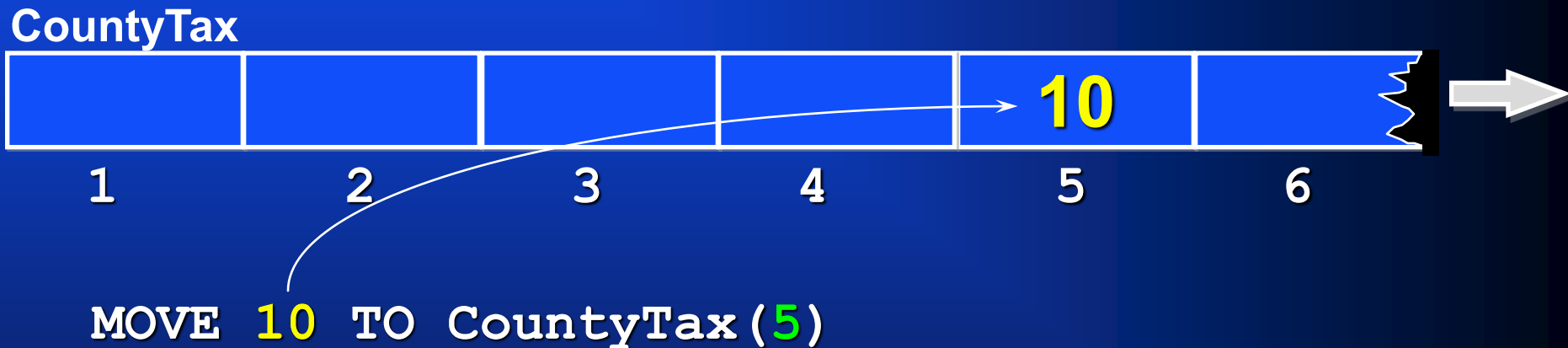
AT END SET EndOfTaxFile TO TRUE

END-READ

58 Statements

Tables/Arrays

A table is a contiguous sequence of memory locations called **elements**, which all have the **same name**, and are uniquely identified by that name and by their **position** in the sequence.

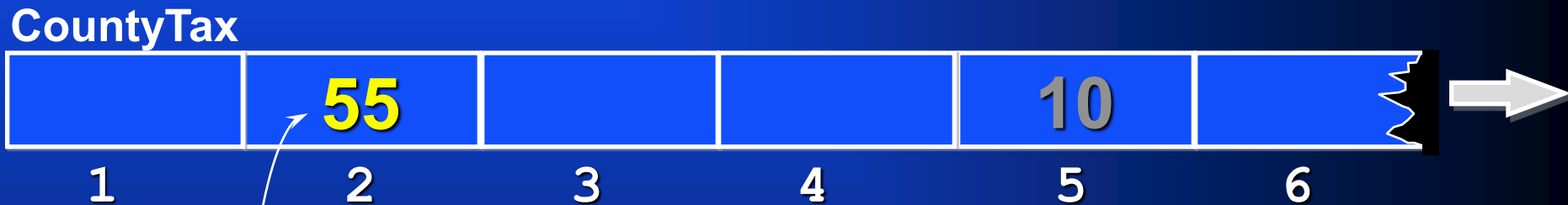


```
ADD TaxPaid TO CountyTax(CountyNum)
```

```
ADD TaxPaid TO CountyTax(CountyNum + 2)
```

Tables/Arrays

A table is a contiguous sequence of memory locations called **elements**, which all have the **same name**, and are uniquely identified by that name and by their **position** in the sequence.



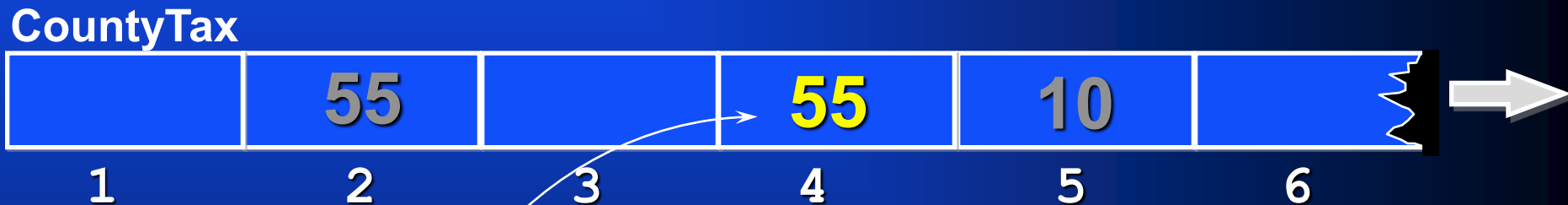
```
MOVE 10 TO CountyTax(5)
```

```
ADD 55TaxPaid TO CountyTax(2CountyNum)
```

```
ADD TaxPaid TO CountyTax(CountyNum + 2)
```

Tables/Arrays

A table is a contiguous sequence of memory locations called **elements**, which all have the **same name**, and are uniquely identified by that name and by their **position** in the sequence.



MOVE 10 TO CountyTax(5)

ADD TaxPaid TO CountyTax(CountyNum)

ADD ⁵⁵TaxPaid TO CountyTax(²CountyNum + 2)

Tables/Arrays

A table is a contiguous sequence of memory locations called **elements**, which all have the **same name**, and are uniquely identified by that name and by their **position** in the sequence. The position index is called a **subscript**.

CountyTax



Subscript

```
MOVE 10 TO CountyTax(5)
```

Subscript points to the value 5 in the above line.

```
ADD TaxPaid TO CountyTax(CountyNum)
```

Subscript points to the variable CountyNum in the above line.

```
ADD TaxPaid TO CountyTax(CountyNum + 2)
```

Subscript points to the expression CountyNum + 2 in the above line.

CountyTax



1

2

3

4

5

6

PROCEDURE DIVISION.

Begin.

OPEN INPUT TaxFile

READ TaxFile

AT END SET EndOfTaxFile TO TRUE

END-READ

PERFORM UNTIL EndOfTaxFile

ADD TaxPaid TO CountyTax(CountyNum)

READ TaxFile

AT END SET EndOfTaxFile TO TRUE

END-READ

END-PERFORM.

PERFORM VARYING Idx FROM 1 BY 1

UNTIL Idx GREATER THAN 26

DISPLAY "County ", CountyNum

" tax total is " CountyTax(Id_x)

END-PERFORM

CLOSE TaxFile

STOP RUN.

Subscript

9 Statements

TaxRecord.

PAYENum

CountyName

TaxPaid

A-89432

CLARE

7894.55

CountyTax



```
IF CountyName = "CARLOW"  
  ADD TaxPaid TO CountyTax(1)  
END-IF
```

```
IF CountyName = "CAVAN"  
  ADD TaxPaid TO CountyTax(2)  
END-IF
```

```
:  
:  
:  
:  
:
```

24 TIMES

TaxRecord.
PAYENum

CountyName

TaxPaid

Idx

A-89432

CLARE

7894.55

1

County

CARLOW

CAVAN

CLARE

CORK

DONEGAL

DUBLIN

1

2

3

4

5

6



CountyTax

500.50

125.75

1000.00

745.55

345.23

123.45

1

2

3

4

5

6



```
PERFORM VARYING Idx FROM 1 BY 1  
    UNTIL County(Idx) = CountyName  
END-PERFORM  
ADD TaxPaid TO CountyTax(Idx)
```

TaxRecord.
PAYENum

CountyName

TaxPaid

Idx

A-89432

CLARE

7894.55

2

County

CARLOW

CAVAN

CLARE

CORK

DONEGAL

DUBLIN

1

2

3

4

5

6



CountyTax

500.50

125.75

1000.00

745.55

345.23

123.45

1

2

3

4

5

6



```
PERFORM VARYING Idx FROM 1 BY 1  
    UNTIL County(Idx) = CountyName  
END-PERFORM  
ADD TaxPaid TO CountyTax(Idx)
```

TaxRecord.
PAYENum

CountyName

TaxPaid

Idx

A-89432

CLARE

7894.55

3

County

CARLOW

CAVAN

CLARE

CORK

DONEGAL

DUBLIN

1

2

3

4

5

6



CountyTax

500.50

125.75

1000.00

745.55

345.23

123.45

1

2

3

4

5

6



```
PERFORM VARYING Idx FROM 1 BY 1  
    UNTIL County(Idx) = CountyName  
END-PERFORM  
ADD TaxPaid TO CountyTax(Idx)
```

TaxRecord.
PAYENum

CountyName

TaxPaid

Idx

A-89432

CLARE

7894.55

3

County

CARLOW

CAVAN

CLARE

CORK

DONEGAL

DUBLIN

1

2

3

4

5

6



CountyTax

500.50

125.75

8894.55

745.55

345.23

123.45

1

2

3

4

5

6



```
PERFORM VARYING Idx FROM 1 BY 1
      UNTIL County(Idx) = CountyName
END-PERFORM
ADD TaxPaid TO CountyTax(Idx)
```

Declaring Tables.

TaxTotals
CountyTax

000000	000000	000000	000000	000000	000000
1	2	3	4	5	6



```
01  TaxTotals.  
    02  CountyTax  PIC 9(10)V99  
                        OCCURS 26 TIMES.
```

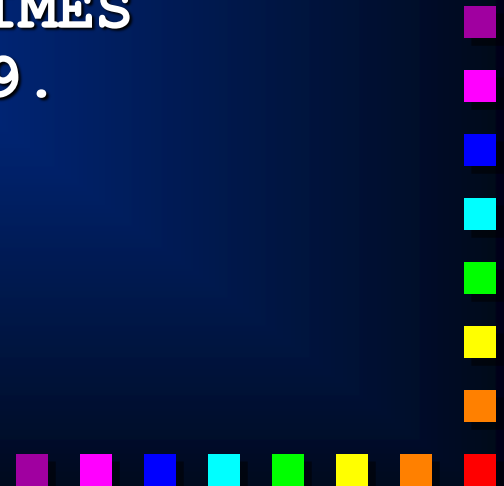
or

```
02  CountyTax  OCCURS 26 TIMES  
                PIC 9(10)V99.
```

e.g.

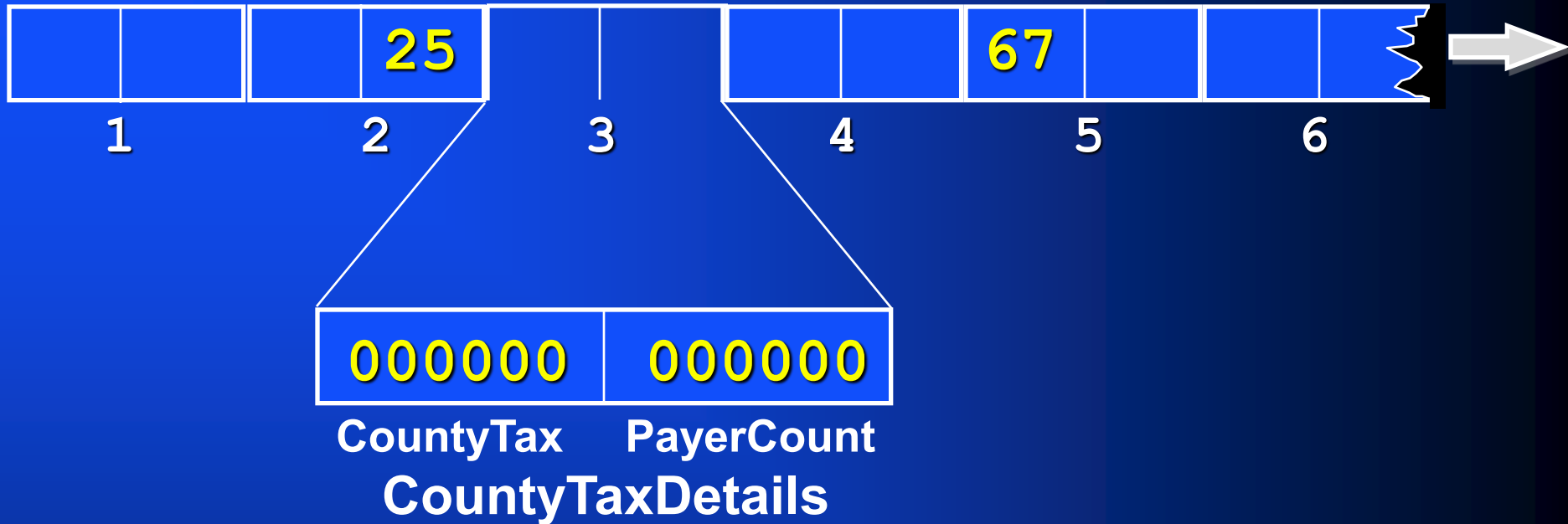
```
MOVE ZEROS TO TaxTotals.
```

```
MOVE 20 TO CountyTax(5).
```



Group Items as Elements.

TaxTotals



```
01  TaxTotals.  
   02  CountyTaxDetails OCCURS 26 TIMES.  
       03  CountyTax      PIC 9(10)V99.  
       03  PayerCount     PIC 9(7).
```

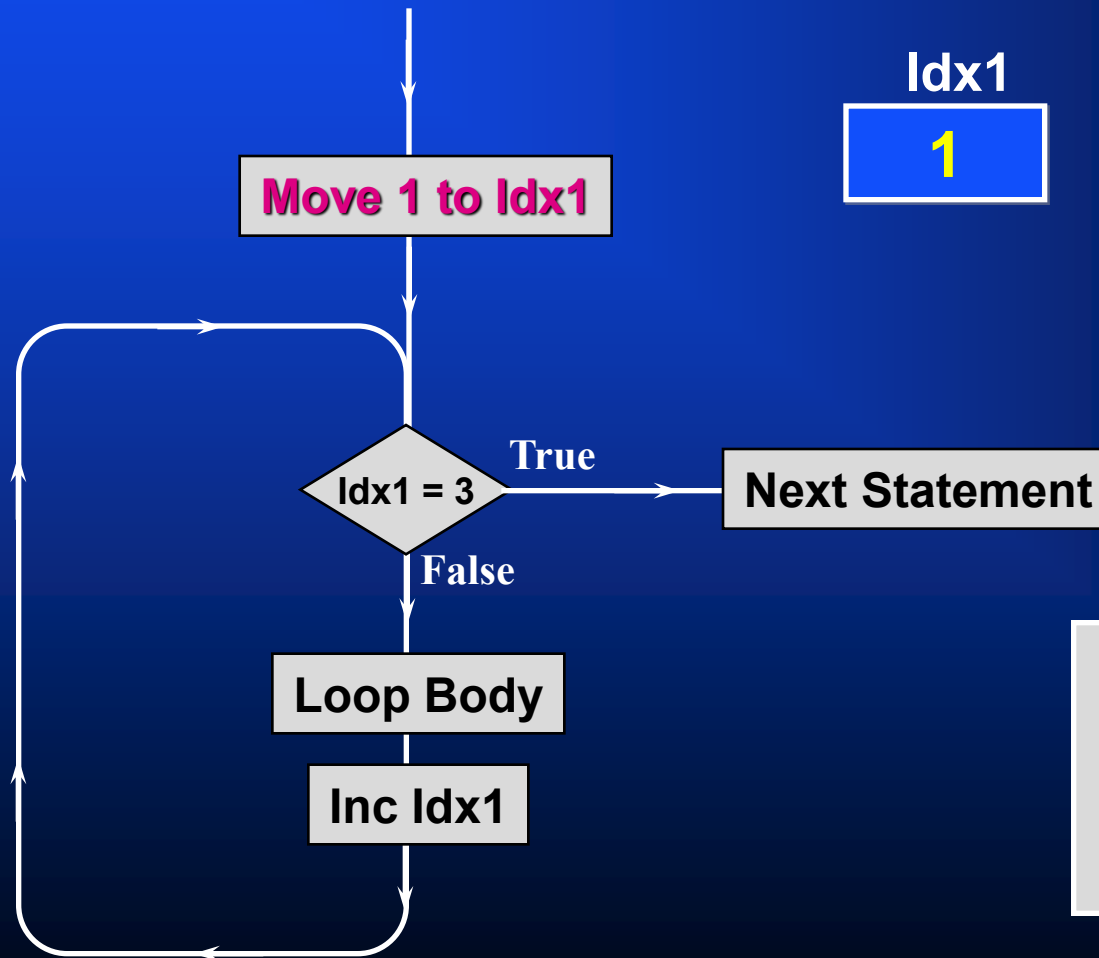
```
e.g.  MOVE 25 TO PayerCount(2).  
       MOVE 67 TO CountyTax(5).  
       MOVE ZEROS TO CountyTaxDetails(3).
```

PERFORM..VARYING Syntax

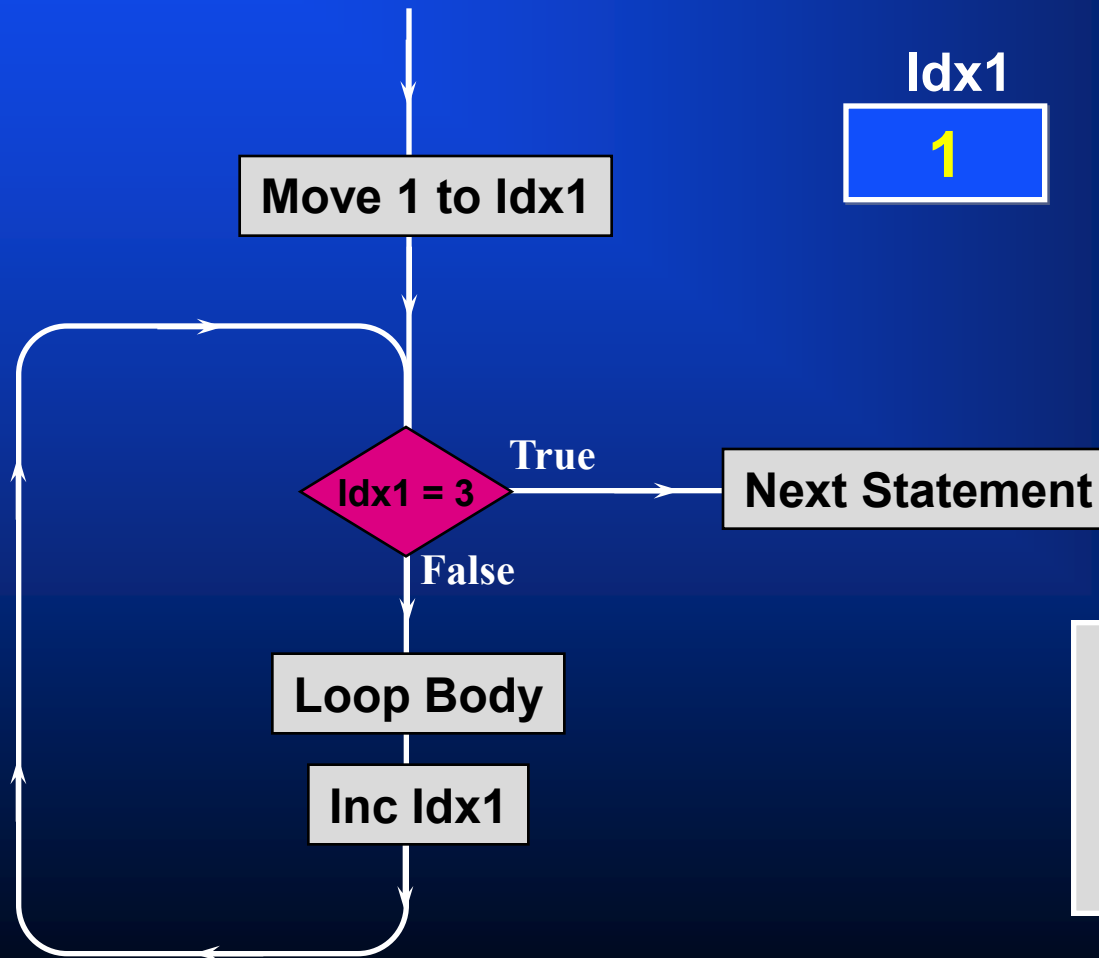
$$\begin{array}{l}
 \underline{\text{PERFORM}} \left[\text{1stProc} \left[\left\{ \begin{array}{c} \underline{\text{THRU}} \\ \underline{\text{THROUGH}} \end{array} \right\} \text{EndProc} \right] \left[\text{WITH } \underline{\text{TEST}} \left\{ \begin{array}{c} \underline{\text{BEFORE}} \\ \underline{\text{AFTER}} \end{array} \right\} \right] \right. \\
 \quad \underline{\text{VARYING}} \left\{ \begin{array}{c} \text{Identifier1} \\ \text{IndexName1} \end{array} \right\} \underline{\text{FROM}} \left\{ \begin{array}{c} \text{Identifier2} \\ \text{IndexName2} \\ \text{Literal} \end{array} \right\} \\
 \quad \underline{\text{BY}} \left\{ \begin{array}{c} \text{Identifier3} \\ \text{Literal} \end{array} \right\} \underline{\text{UNTIL}} \text{Condition1} \\
 \quad \left[\begin{array}{c} \underline{\text{AFTER}} \left\{ \begin{array}{c} \text{Identifier4} \\ \text{IndexName3} \end{array} \right\} \underline{\text{FROM}} \left\{ \begin{array}{c} \text{Identifier5} \\ \text{IndexName4} \\ \text{Literal} \end{array} \right\} \\
 \quad \quad \underline{\text{BY}} \left\{ \begin{array}{c} \text{Identifier6} \\ \text{Literal} \end{array} \right\} \underline{\text{UNTIL}} \text{Condition2} \end{array} \right] \dots \\
 \left[\text{StatementBlock } \underline{\text{END - PERFORM}} \right]
 \end{array}$$



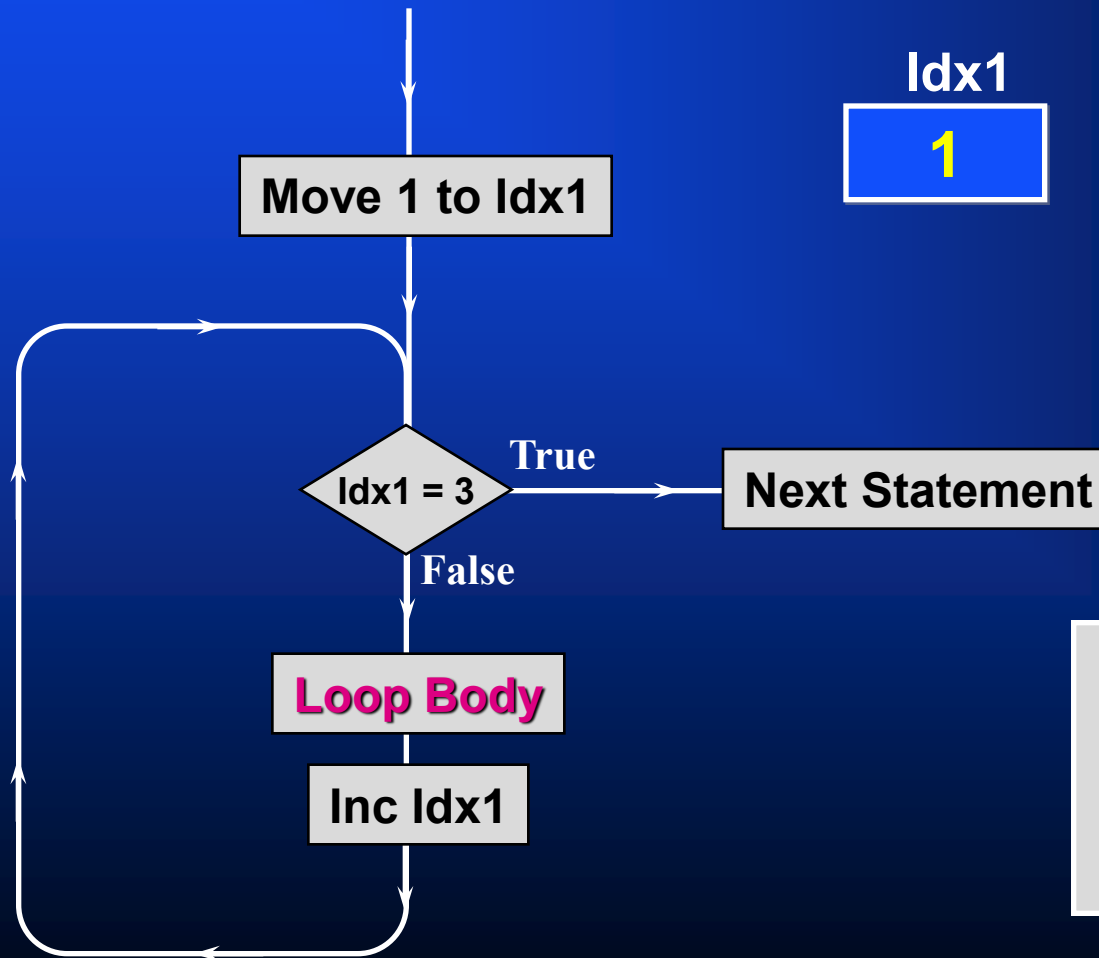
```
PERFORM VARYING Idx1 FROM 1 BY 1 UNTIL  
    Idx1 EQUAL TO 3  
    DISPLAY Idx1  
END-PERFORM.
```



```
PERFORM VARYING Idx1 FROM 1 BY 1 UNTIL  
    Idx1 EQUAL TO 3  
    DISPLAY Idx1  
END-PERFORM.
```



```
PERFORM VARYING Idx1 FROM 1 BY 1 UNTIL  
    Idx1 EQUAL TO 3  
    DISPLAY Idx1  
END-PERFORM.
```

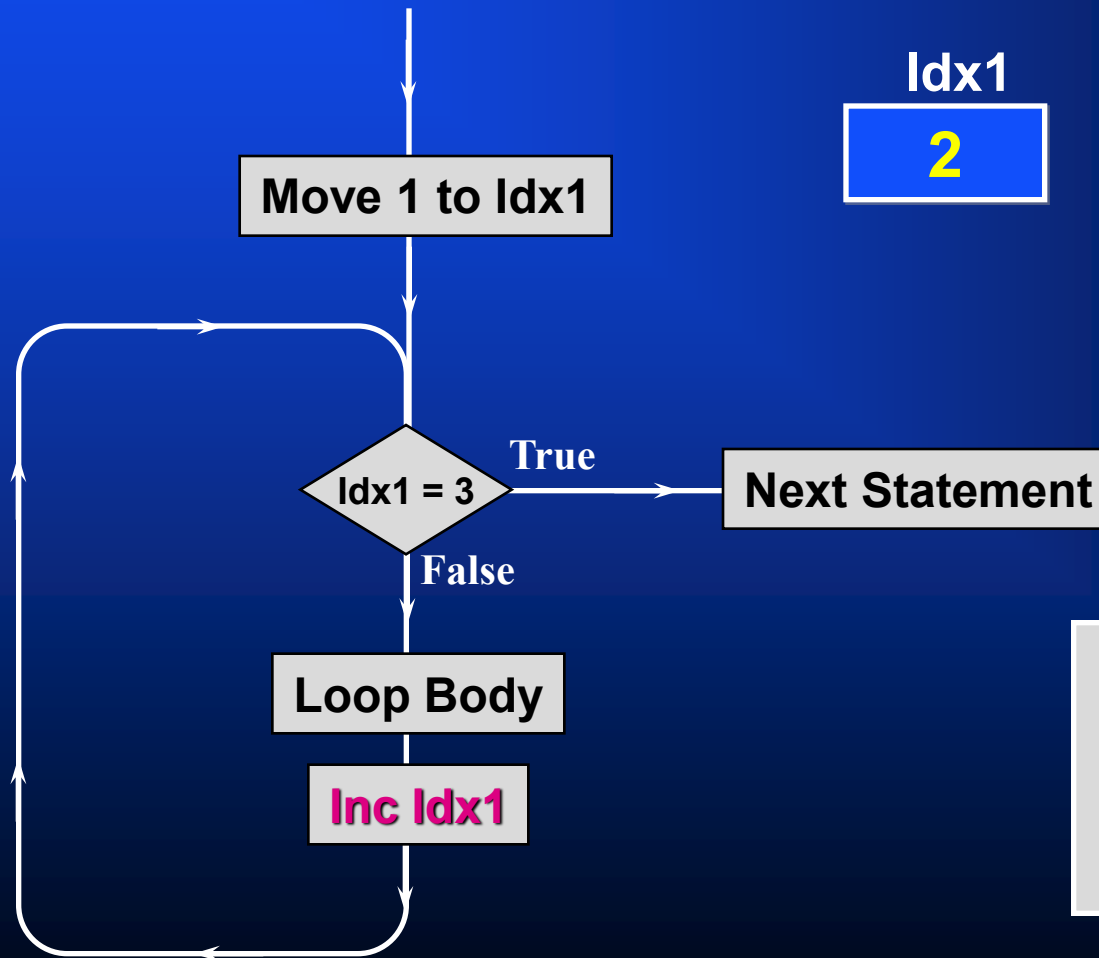


Idx1

1

1

```
PERFORM VARYING Idx1 FROM 1 BY 1 UNTIL  
    Idx1 EQUAL TO 3  
    DISPLAY Idx1  
END-PERFORM.
```

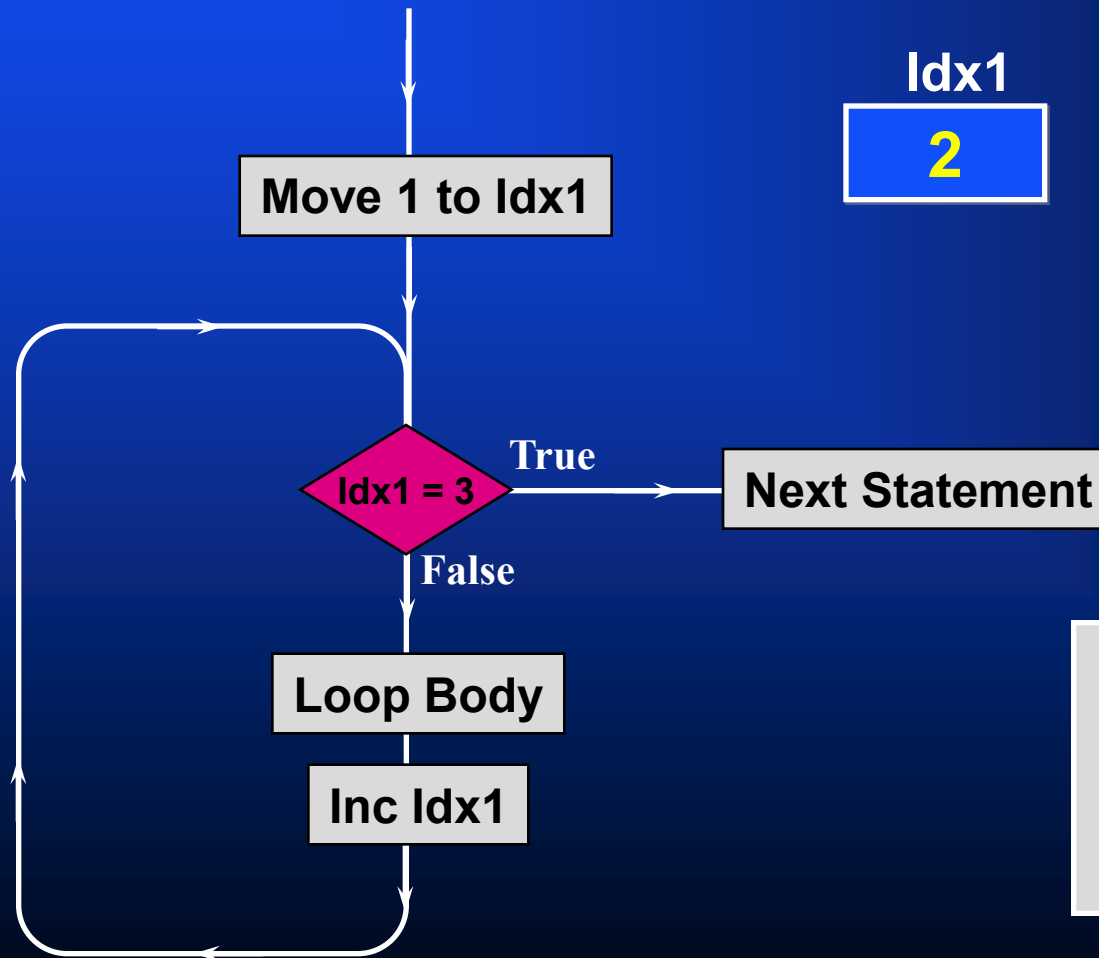


Idx1

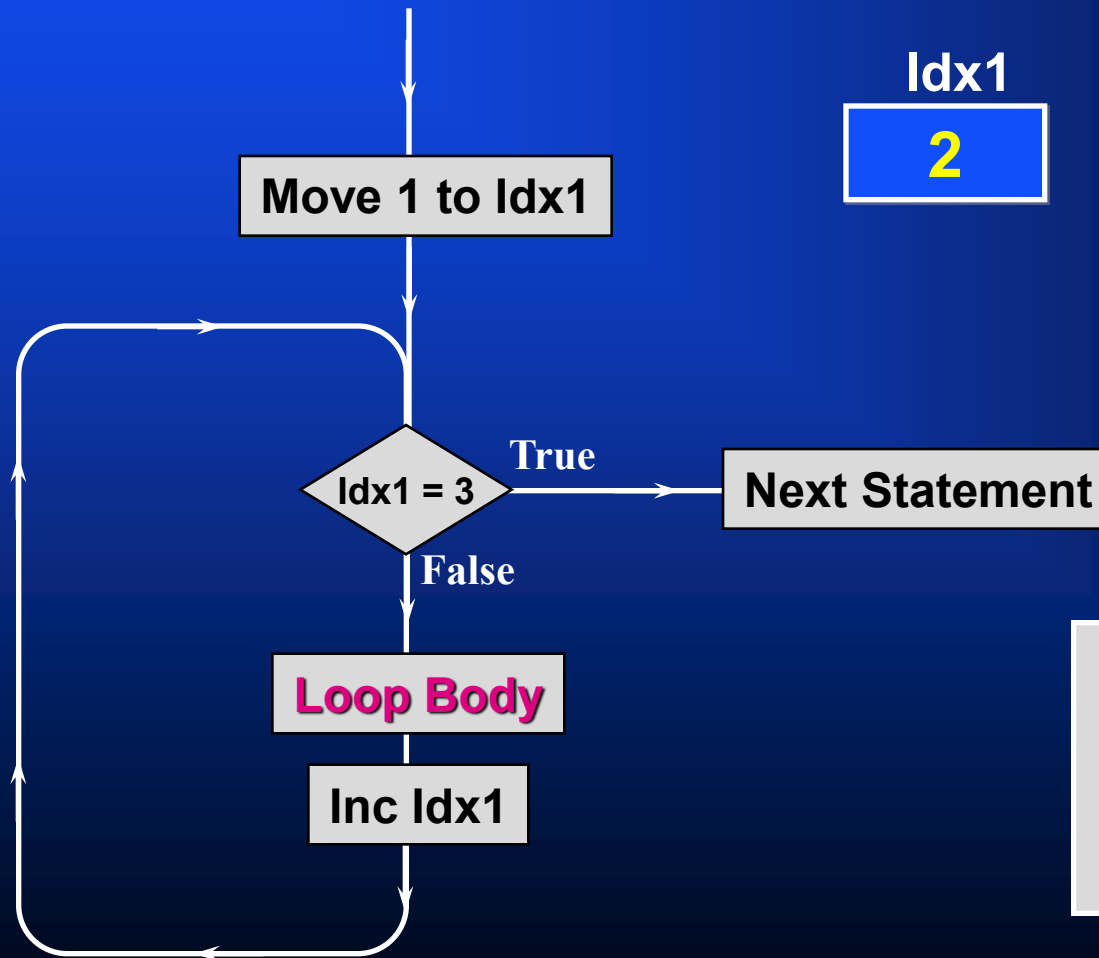
2

1

```
PERFORM VARYING Idx1 FROM 1 BY 1 UNTIL  
    Idx1 EQUAL TO 3  
    DISPLAY Idx1  
END-PERFORM.
```



```
PERFORM VARYING Idx1 FROM 1 BY 1 UNTIL  
    Idx1 EQUAL TO 3  
    DISPLAY Idx1  
END-PERFORM.
```



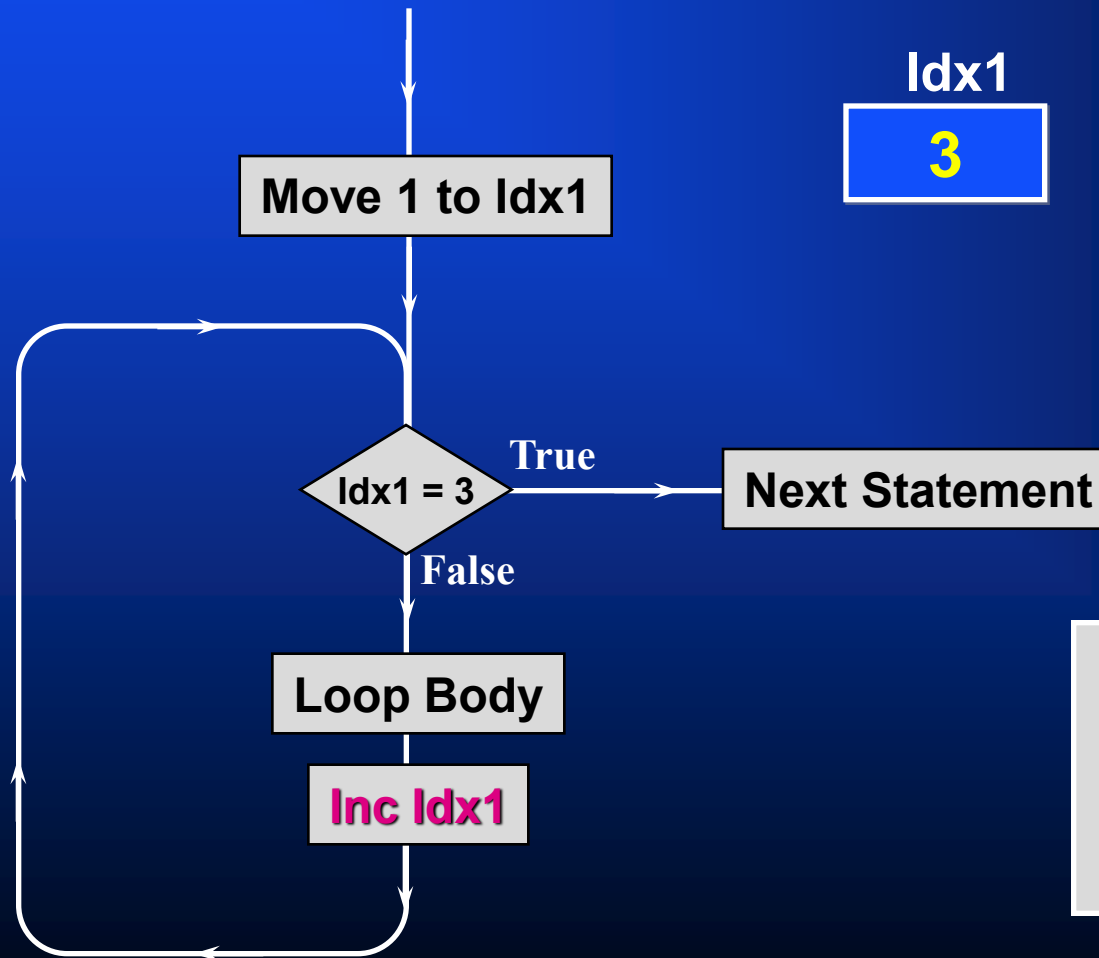
Idx1

2

1

2

```
PERFORM VARYING Idx1 FROM 1 BY 1 UNTIL  
    Idx1 EQUAL TO 3  
    DISPLAY Idx1  
END-PERFORM.
```



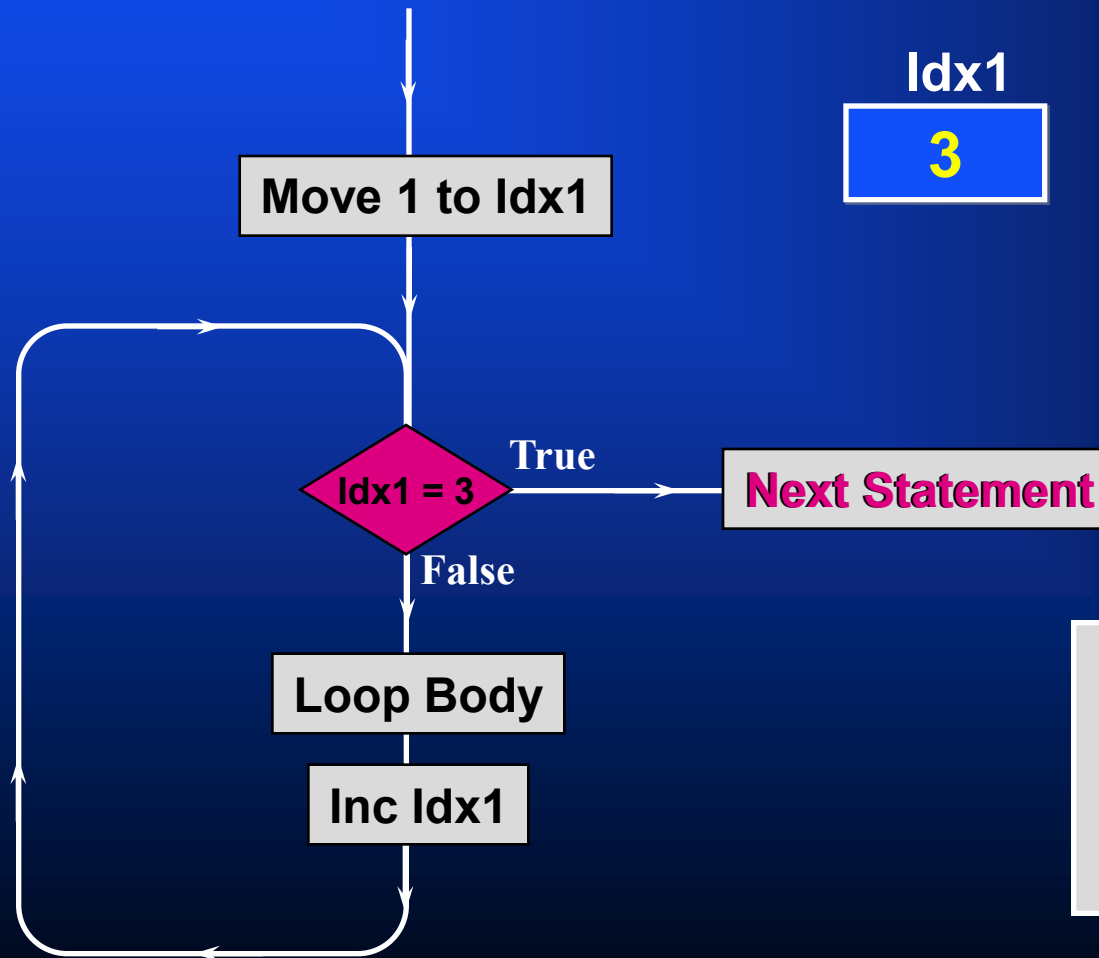
Idx1

3

1

2

```
PERFORM VARYING Idx1 FROM 1 BY 1 UNTIL  
    Idx1 EQUAL TO 3  
    DISPLAY Idx1  
END-PERFORM.
```



Idx1

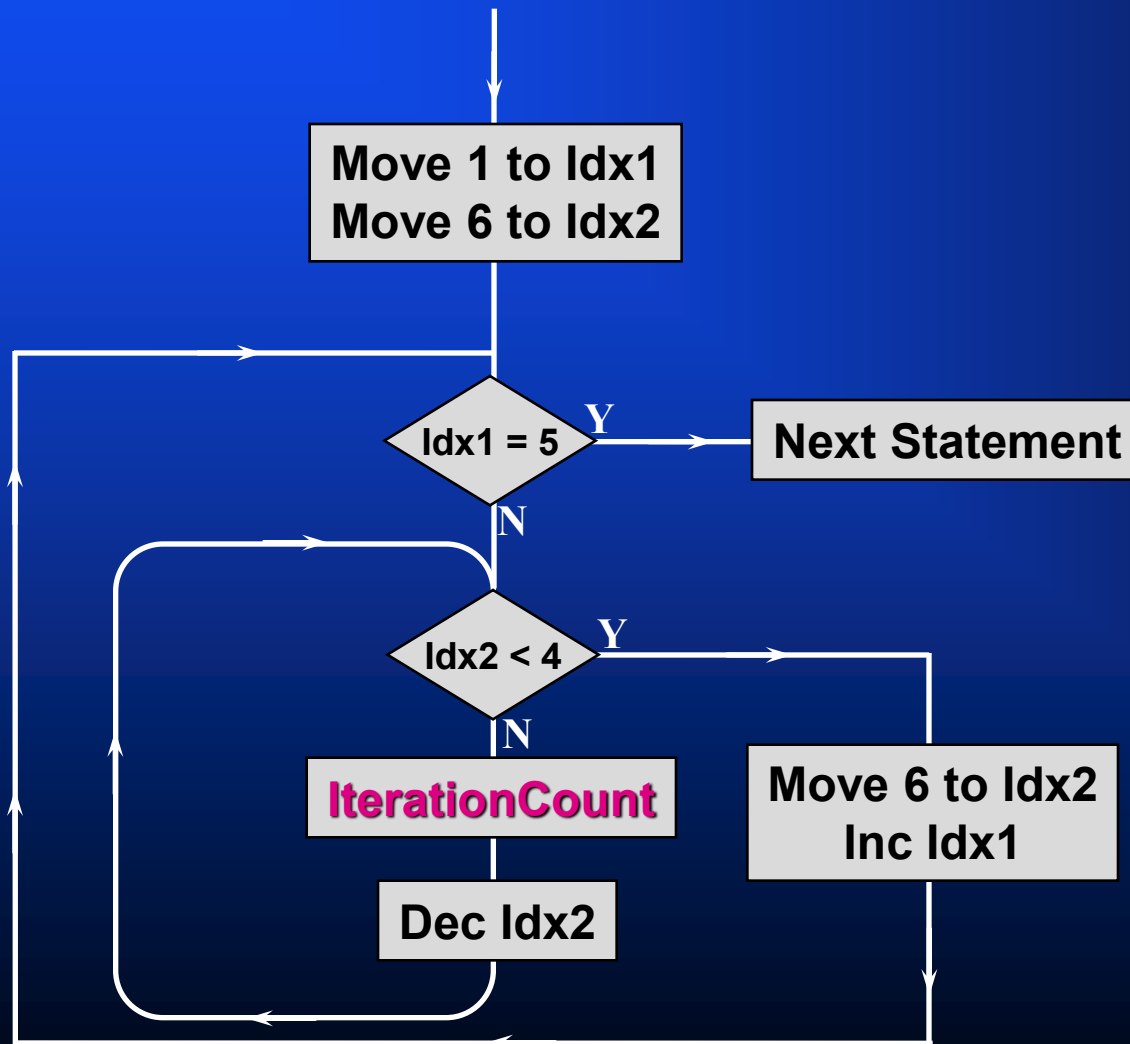
3

1

2

Exit value = 3

PERFORM IterationCount VARYING Idx1 FROM 1 BY 2
UNTIL Idx1 EQUAL TO 5
AFTER Idx2 FROM 6 BY -1
UNTIL Idx2 LESS THAN 4



T	Idx1	Idx2
1	1	6
2	1	5
3	1	4
4	3	6
5	3	5
6	3	4
x	= 5	= 6